

Exploratory Data Analysis of the Real Estate Market

This project performs Exploratory Data Analysis (EDA) on a real estate market dataset containing housing prices, rental prices, mortgage rates, and district-level information. The objective of the project is to identify trends, patterns, correlations, and anomalies within the housing market using statistical analysis and visualization techniques.

Attribute	Value
Rows	9804
Columns	13
Dataset Type	Real Estate Market Dataset

Objectives

- Understand dataset structure and variables
- Analyze missing values and data quality issues
- Identify trends and patterns in housing prices
- Explore correlations between economic indicators and property prices
- Detect outliers and anomalies
- Perform district-wise market analysis
- Conduct statistical hypothesis testing

Key Findings

- Property prices increased steadily from 2020 to 2026.
- Newbuild properties are generally more expensive than secondary properties.
- Strong positive correlations exist among property price variables.
- Mortgage rates closely follow central bank key rates.
- Premium districts exhibit significantly higher housing prices.
- Luxury properties create visible outliers in price distributions.

Visualizations Included

- Property prices over time
- Missing Value Heatmap
- Correlation Heatmap
- Histograms
- Boxplots
- Time-Series Trend Analysis
- District-Wise Analysis

Hypothesis Testing

A two-sample t-test was conducted to compare secondary property prices and newbuild property prices. The p-value obtained from the test was significantly below 0.05, indicating a statistically significant difference between the two

property categories.

Conclusion

The exploratory data analysis successfully identified major trends, relationships, and geographic variations within the housing market dataset. The project demonstrated the usefulness of statistical analysis and data visualization techniques in understanding real-world housing market behavior.

Future Scope

- Predictive modeling of housing prices
- Time-series forecasting
- Machine learning applications
- Geographic clustering analysis
- Inflation-adjusted housing market analysis

Prepared By: Dipesh Dubey